ME 5309 – Mid-Semester Examination: ANSWER SHEET

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Questions attempted: Q1, Q2, Q3

Question 1

(a) (i) $f_x = f/p = 16\text{ mm} / 5\ \mu\text{m} =$ **3200 px**.

(ii) Sensor width = $1280 \times 5\ \mu\text{m} = 6.4$ mm; FOV = $W_s \cdot Z/f = 6.4 \times 400/16 =$ **160 mm**.

(b) Assumptions: ideal 12-bit ADC (4096 codes), the full 0–5 V range is used, no offset or gain error, and sensor noise is below one LSB so quantisaton sets the limit.

Voltage step: 1 LSB = $V_{FS}/2^N = 5\text{ V}/4096 = 1.2207\times10^{-3}$ V = 1.2207 mV.

The sensor maps force to voltage by $V = S\cdot F$ with S = 10 mV/N, so $\Delta F_{min} = \Delta V/S = 1.2207$ mV / (10 mV/N) = 0.12207 N. $\mathbf{\Delta F_{min} \approx 0.122\ N}$. (Using 4095 steps instead of 4096 codes gives 0.1221 N, the same at 3 s.f.)

(c) ZOH discretisaton: $F = e^{AT}$, $G = \int_0^T e^{A\tau}B\,d\tau = A^{-1}(F - I)B$ $(A \neq 0)$.

Here $A = -2$, $B = 1$, $T = 0.1$: $F = e^{-0.2} =$ **0.8187**. $G = (F - 1)/A = (0.8187 - 1)/(-2) =$ **0.0906**.

$x_{k+1} = 0.8187\ x_k + 0.0906\ u_k$.

Question 2

(a) The model is second order, so I choose the position and velocity as states: $x = [x_1, x_2]^T = [\theta, \dot{\theta}]^T$, with input τ.

Rewriting $J\ddot{\theta} + b\dot{\theta} + mg\ell \sin\theta = \tau$ gives the nonlinear state-space model: $\dot{x}_1 = x_2$, $\dot{x}_2 = (1/J)(\tau - b\,x_2 - mg\ell \sin x_1)$, i.e. $\dot{x} = f(x, \tau)$.

Equilibria for constant τ_0 need $\dot{x} = 0$: the first equation gives $x_2 = 0$, and the second then gives $mg\ell \sin\theta_e = \tau_0$, i.e. $\sin\theta_e = \tau_0/(mg\ell)$.

This has a solution only if $|\tau_0/(mg\ell)| \leq 1$, so $\mathbf{|\tau_0| \leq mg\ell = 2.4525\ N\cdot m}$. For $|\tau_0| < mg\ell$ there are two equilibria per revolution (θ_e and $\pi - \theta_e$); at $|\tau_0| = mg\ell$ they merge, and beyond it the joint cannot be held statically.

(b) $\delta\dot{x} = A\,\delta x + B\,\delta\tau$, $A = \partial f/\partial x$, $B = \partial f/\partial\tau$ at $\theta_e = 60°$, $x_{2e} = 0$.

$A = [0, 1;\ -(mg\ell/J)\cos\theta_e, -b/J] = [0, 1;\ -122.6\times0.5, -2.5] =$ **[0, 1; −61.31, −2.5]**.

$B = [0; 1/J] =$ **[0; 50]**.

(c) At rest $\dot{\theta} = \ddot{\theta} = 0$, so $K_p(\theta_d - \theta_{ss}) = mg\ell \sin\theta_{ss}$, i.e. $50(90 - \theta_{ss}) = 2.4525 \sin\theta_{ss}$.

Gravity torqe is non-zero at 90°, so the left side can't be zero, hence $\theta_{ss} \neq \theta_d$.

Taking $\sin\theta_{ss} \approx 1$: $90 - \theta_{ss} = 2.4525/50 = 0.049$, so the stedy-state error ≈ **0.049°**.

(d) Steady state: $\dot{\theta} = \ddot{\theta} = 0$ and the control law gives $\tau = K_p(\theta_d - \theta_{ss}) + mg\ell \sin\theta_{ss}$. Plugging into the plant, $mg\ell \sin\theta_{ss} = K_p(\theta_d - \theta_{ss}) + mg\ell \sin\theta_{ss}$, so $K_p(\theta_d - \theta_{ss}) = 0$ and $\mathbf{\theta_{ss} = \theta_d}$: **stedy-state error is zero**.

Poles: linearise the closed loop about θ_d. Before compensation, $J\ddot{\theta} + (b + K_d)\dot{\theta} + (K_p + mg\ell\cos\theta_d)\,\delta\theta = 0$, characteristic polynomial $J\lambda^2 + (b + K_d)\lambda + K_p + mg\ell\cos\theta_d$. After compensation the added term $mg\ell \sin\theta$ cancels the plant's gravity term exactly, leaving $J\lambda^2 + (b + K_d)\lambda + K_p$.

The two differ only by $mg\ell\cos\theta_d$, and for $\theta_d = 90°$ this is $2.4525 \times 0 = 0$. So **the closed-loop poles are unchanged** at this operating angle. I should stress this is because $\cos 90° = 0$; if θ_d were, say, 60°, the compensation would raise the stiffness by 61.3×0.02... i.e. it would move the pols.

Question 3

(a) Reason 1 – noise amplification: the backward difference $\hat{\omega}_k = (\theta_k - \theta_{k-1})/T$ has frequncy response $(1 - e^{-j\omega T})/T$, whose magnitude grows with frequncy ($\approx \omega$ for small ωT), i.e. it is a high-pass operation. Any noise on the encoder signal therefore appears in the velocity multiplied by roughly $1/T = 1000\ s^{-1}$, and can swamp the true velocity.

Reason 2 – quantisation: the encoder only reports integer counts, so $\Delta\theta$ is an integer multiple of the count size q. Dividing by T = 1 ms gives velocity steps of q/T; for a 4096-count encoder $q = 2\pi/4096 = 1.53\times10^{-3}$ rad, so one step is 1.53 rad/s. At low speed the estimate flips between 0 and one step (a noisy staircase), exactly where the damping term of the PD law is needed.

(A further, smaller issue is the half-sample delay of a backward difference, but the two reasons above are the main ones.)

(b) $H(s) = \omega_c/(s + \omega_c)$, $\omega_c = 2\pi(20) = 125.66$ rad/s. Tustin: $s = (2/T)(z - 1)/(z + 1)$, T = 1 ms. With $a = \omega_c T/2 = 0.06283$: $H(z) = a(1 + z^{-1}) / [(1 + a) - (1 - a)z^{-1}]$.

$y_k = [a/(1+a)](u_k + u_{k-1}) + [(1-a)/(1+a)]\, y_{k-1}$ = **$0.05912\,(u_k + u_{k-1}) + 0.88177\, y_{k-1}$**.
DC gain: z = 1 gives $H(1) = 2(0.05912)/(1 - 0.88177) = 0.11824/0.11823 = 1.000$ ✓.

(c) $A - LC = [-\ell_1, 1; -\ell_2, -2]$, so $\det(sI - A + LC) = (s + \ell_1)(s + 2) + \ell_2 = s^2 + (2 + \ell_1)s + (2\ell_1 + \ell_2)$.
Desiered: $(s + 10)(s + 10) = s^2 + 10s + 100$. Matching: $2 + \ell_1 = 10 \Rightarrow \ell_1 = 8$; $2(8) + \ell_2 = 100 \Rightarrow \ell_2 = 84$.

$L = [8, 84]^T$.

(d) Advantage of the observer that no filter of this kind can give: the observer is driven by both the measurement y and the model with the actual input u, so it predicts the motion between samples. The velocity estimate therefore has little or no phase lag. The low-pass fliter on differenced data, by contrast, always pays a lag of about $1/(2\pi f_c) = 8$ ms for its noise rejection, a trade-off it cannot escape. In addition the observer gives all states, and can be augmented with a disturbance state (e.g. unknown load torqe) which no fliter on θ can estimat.

Circumstance where the filter is safer: when the plant model is poorly known or changes during operation, for example an unknown payload, varying friction, contact with the environment, or actuator saturation making u differ from the commanded value. The obsever propagates the wrong model so its estimate can be biased (and in the worst case unstable with the feedback loop), whereas the model-free filter just smooths what the encoder actually measures.